

Judge Brief

RetiNova — AI-assisted diabetic retinopathy screening · SIH25131

Everything the panel is likely to ask, with a two-line answer for the room and the detail behind it.

Every measured figure was produced by the running system and can be reproduced from the repository.

1. The dataset — asked first, almost every time

Where the model was trained

Source	HuggingFace — yousefedweqd/Diabetic_Retinopathy_Detection huggingface.co/datasets/yousefedweqd/Diabetic_Retinopathy_Detection
Licence	MIT — permissive, commercial use allowed
Size	35,126 images — 25,290 train / 2,810 validation / 7,026 test (1.35 GB, parquet)
Labels	0–4 on the ICDR severity scale, one grade per image
Balance	Heavily skewed: 18,583 grade 0 against 509 grade 4 in the training split
True origin	Almost certainly EyePACS — the Kaggle <i>Diabetic Retinopathy Detection</i> (2015) set, whose training split is exactly 35,126 images with the same 0–4 labels. The re-host states no provenance, so this is inference, not documentation — say so if asked.

Where it was validated — datasets the model never saw

Dataset	Source	Licence	Images	Why this one
IDRiD	amin-nejad/idrid-disease-grading	CC-BY-4.0	516	Indian population , Kowa VX-10a camera, collected in Nanded, Maharashtra
Messidor-2	sngsfydy/Messidor2	not stated	1,744	French population, Topcon TRC NW6 — the standard DR benchmark

The line to use: "Trained on 35,126 EyePACS-derived images under MIT licence, and validated without retraining on two independent datasets — including 516 Indian images from a different camera, where it reached 95.4% sensitivity."

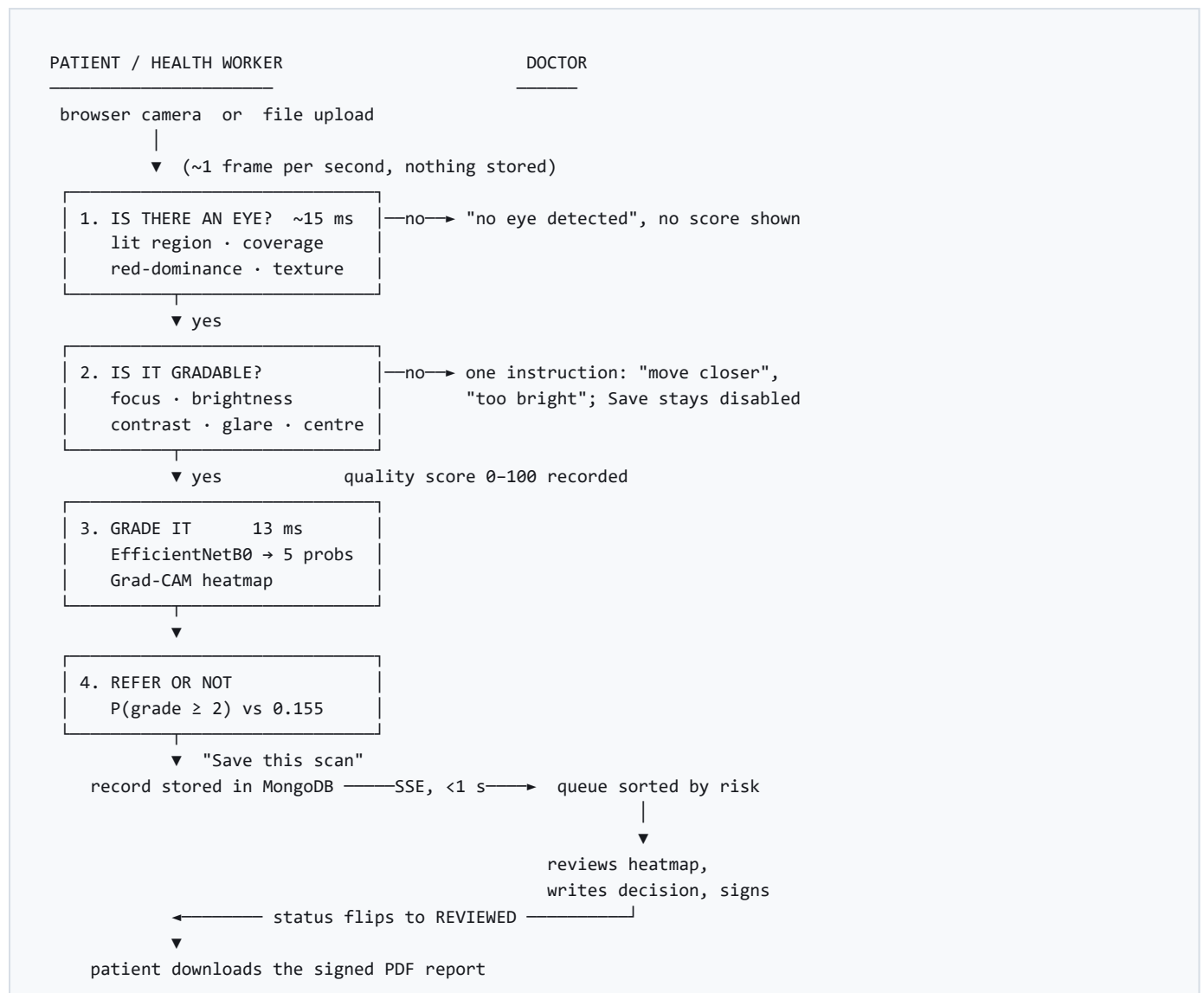
2. The technology, block by block

Block	Technology	Why this and not something else
Web app	React 19.2 + Vite 6, react-router 7	The camera, the live loop and the report all run in the browser, so no install and any phone works. Vite builds it to static files that cost nothing to host.
API / business logic	Spring Boot 4.1, Java 17	Auth, roles, records, the doctor workflow. Java because it is what the team knows and what hospital IT departments already run.
AI service	FastAPI + ONNX Runtime + OpenCV	Kept separate so the model can be retrained and redeployed without touching the clinical workflow. ONNX, not TensorFlow: 16 MB instead of ~600 MB, 13 ms per image, and it fits in a free container.
Database	MongoDB Atlas	A screening is one document — image, grade, probabilities, quality, review — and its shape changed repeatedly during the build. Images live in the record because container disks are wiped on every deploy.
Auth	JWT (jjwt) + BCrypt	Stateless tokens, so the three services scale independently. Passwords are hashed, never stored.
Live updates	Server-Sent Events	A new screening appears on the doctor's queue in under a second with no refresh. SSE, not WebSockets: the traffic is one-way, so the simpler tool is the right one.
Training	TensorFlow / Keras, EfficientNetB0	Training only, never shipped. EfficientNetB0 has 5.3 M parameters — the accuracy of far larger networks at a size that runs on a CPU.
Explainability	Grad-CAM, computed as CAM	For a Global-Average-Pooling head, Grad-CAM's gradients equal the final layer's weights, so the heatmap needs one forward pass and no autodiff. Verified: probabilities match to 1e-6, heatmaps correlate at 0.97.
Quality gate	Classical OpenCV	Deliberately not a second neural network — every threshold is inspectable, needs no training data, and costs ~1 ms, which is what lets it run on every live frame.
Hosting	Render (3 services) + MongoDB Atlas	Free tier throughout. Docker for the two backends, static CDN for the frontend.

The three deployed services

Service	Type	Responsibility
retinova / frontend	static site	Everything the user sees; camera capture; live loop; printable report
dr-backend	docker	Auth, patients, screenings, triage, doctor queue, SSE, image serving
dr-ai-service	docker	Quality gate, grading, heatmap, referral decision, the three triage signals

3. How it works end to end



4. What it does — the feature list

Feature	What it actually does
Retinal grading	Five ICDR grades with the full probability spread, not just the winner
Image quality gate	Refuses non-eyes in ~15 ms and ungradable eyes with a reason; records a 0–100 score on every screening
Live capture guidance	Target ring plus one instruction at a time; Save disabled until the frame is usable
Referral decision	A threshold on $P(\text{grade} \geq 2)$ chosen for 80% sensitivity, printed with what it costs
Explainability	Grad-CAM heatmap and overlay stored with every record
Camera-only triage	Pulse (PPG), pupil light reflex (PLR), conjunctival pallor → a priority score with no extra hardware
Laterality	Which eye (OD / OS) recorded per screening
Doctor workflow	Live queue, review, decision, remarks, signature; nothing finalises without a human
Printable report	Patient identity, grade, images, meaning, next step, limitations, reviewer — vector PDF via print CSS
History	Every screening and triage kept per patient with images and timestamps
Security	JWT auth, BCrypt passwords, role separation, ownership-checked image URLs

5. Technical questions the panel will ask

Q1. Why EfficientNetB0 and not ResNet50, VGG16, or a transformer?

It reaches comparable accuracy with 5.3 million parameters instead of 25–138 million, which is what lets the whole service run on a free CPU container in 13 ms.

In detail: EfficientNet scales depth, width and resolution together rather than one at a time, so it sits on a better accuracy-per-parameter curve. A vision transformer would likely beat it given ImageNet-21k pretraining and a GPU — we have neither, and a model that cannot be served is not a product. The architecture is a swap of one line if that changes.

Q2. Your accuracy is 44%. Isn't that terrible?

Accuracy is the wrong measure here and we do not use it. The system makes one decision — refer or not — and on that it is right 81% of the time on referable cases.

In detail: 73% of the data is grade 0, so a model that always answered "No DR" would score 73% accuracy and be worthless. Screening is judged on sensitivity, specificity and quadratic weighted kappa. Ours: **kappa 0.425, sensitivity 80.9%, specificity 60.3%** on 7,026 held-out images — and 95.4% sensitivity on an Indian dataset it had never seen.

Q3. What is quadratic weighted kappa and why use it?

It measures agreement with human graders after subtracting the agreement you would get by chance, and it punishes big mistakes far more than small ones.

In detail: The grades are ordered, so calling a grade-4 eye "grade 0" is much worse than calling it "grade 3". Quadratic weighting penalises by the square of the distance. 1.0 is perfect, 0 is chance. Two human graders typically agree around 0.80; ours is 0.425 internally and 0.737 on IDRiD.

Q4. Why is your specificity so low? Won't you flood the clinic?

It is low on purpose. We chose the threshold to catch 4 in 5 referable cases, and that choice costs us 4 in 10 healthy people being referred.

In detail: Sensitivity and specificity trade against each other along one curve. Using the model's most likely grade gives 89% specificity but only 50% sensitivity — half the sick people sent home. For screening, a missed case costs sight; a false alarm costs an appointment. The threshold is a single number in a config file, so a programme that wants the other balance can move it — and the report prints whichever pair is in force.

Q5. How do you know the model isn't just memorising this dataset?

We ran it, unchanged, on two datasets from other countries and other cameras. Sensitivity held between 81% and 95% with no retraining.

In detail: On IDRiD (Indian, Kowa camera): 95.4% sensitivity, 65.3% specificity, kappa 0.737. On Messidor-2 (French, Topcon): 82.9% and 51.4%, kappa 0.549. The honest caveat from the same experiment: per-grade accuracy does **not** transfer — Moderate recall is 7% on IDRiD while Severe is 87%, so the model pushes middle grades outward and they land on the referable side. The referral decision generalises; the five-grade label does not.

Q6. Explain Grad-CAM. Is it real explainability or decoration?

It shows which parts of the retina moved the prediction, by weighting the final feature maps by how much each contributed to the chosen grade.

In detail: Our head is Global Average Pooling → Dense, and for that shape the Grad-CAM gradient weights are algebraically identical to the Dense weights — so the heatmap is a single forward pass, no autodiff, no TensorFlow at serving time. It is genuine attribution, not decoration, but it is coarse: it points at a region, not at a lesion. We present it as "where it looked", never as "this is the haemorrhage".

Q7. Why ONNX instead of just serving the Keras model?

The TensorFlow image was around 600 MB and would not fit the free tier; the ONNX model is 16 MB and answers in 13 ms.

In detail: ONNX (Open Neural Network Exchange) is a framework-independent format. We train in Keras, export once, and serve with ONNX Runtime. The export script refuses to write anything unless the two agree to within $1e-5$ on random input; measured agreement is $8.6e-07$.

Q8. Why microservices instead of one application?

The model changes on a different clock from the clinical workflow, and the AI service is the only part that needs heavy Python libraries.

In detail: Retraining and redeploying the model must not risk the auth, records or doctor review path. Splitting also lets the AI service scale on its own, and keeps a 600 MB ML dependency tree out of the service that holds patient data. The cost is network hops and three cold starts, which is the right trade here.

Q9. How is patient data protected?

JWT authentication, BCrypt-hashed passwords, role separation, and every image URL checks that the requester owns that record.

In detail: Tokens expire; passwords are never stored, only their BCrypt hash; a patient cannot open another patient's screening or a doctor-only route. Images are served through an ownership-checked endpoint rather than a public path. **What is missing, honestly:** encryption at rest is Atlas's default rather than something we configured, there is no rate limiting, and no audit log of who viewed what. Those are required before real patient data, and they are on the list.

Q10. What happens if the AI service goes down?

The health endpoint reports it, screening is blocked with an explanation instead of a failure, and everything already stored stays readable.

In detail: `/api/health` reports each of the three services independently. Grading is the only feature that hard-depends on the AI service; login, history, reports and doctor review continue. On the free tier a sleeping service takes ~50 s to wake, which the UI states plainly rather than hanging.

Q11. Can this run offline, in a village with no internet?

Not today. The architecture supports it — there is a docker-compose file — but we have not built or tested an offline deployment, so we do not claim it.

In detail: All three services plus MongoDB can run on one laptop on a local network; the model is 16 MB and the inference is CPU-only, so there is no technical obstacle. What is missing is a tested build, a sync path for when connectivity returns, and a way to update the model. Roughly half a day of work, not a redesign.

Q12. How long does one screening take?

13 ms of model time; about one second end to end locally, and 1–2 seconds through the free cloud tier.

In detail: Rejecting a non-eye takes 11–23 ms because the model never runs — roughly fifty times cheaper than grading, which is exactly why that check can run on every live frame. The rest is JPEG encoding, network and heatmap rendering.

Q13. Your training data is 73% healthy. How did you handle that?

Class weights during training, keeping every image of the rare grades, and judging the model on sensitivity rather than accuracy.

In detail: Weights are square-rooted inverse frequency — full inverse weighting makes one grade-4 image worth eight grade-0 images and turns the gradient into noise. The training set keeps all of grades 1–4 and caps only grade 0. An earlier version capped every class equally, which threw away 1,300 grade-2 images while doing nothing for grade 4.

Q14. What is the false negative rate, and what happens to those people?

About 1 in 5 referable cases is missed at our threshold. They are told to attend routine annual screening, so they are seen again — later than ideal, not never.

In detail: This is the number that matters most and we print it on every report. It is why the system is positioned as triage rather than diagnosis, why a doctor reviews every case, and why a low grade is never phrased as "you are fine". Moving the threshold to 90% sensitivity is one config change and costs specificity — a programme-level decision, not ours.

Q15. Why did you not use APTOS, the more famous dataset?

APTOS requires accepting Kaggle competition terms interactively, which cannot be automated. We used an MIT-licensed set of double the size instead.

In detail: 35,126 images against APTOS's 3,662. We did search for and verify ungated mirrors of APTOS, IDRiD, Messidor-2 and DDR, and used IDRiD and Messidor-2 for external validation — which is the more valuable role for a second dataset anyway.

Q16. How would you improve it with more time and a GPU?

An ordinal head, higher input resolution, and lesion-level labels — in that order.

In detail: (1) The head predicts five independent classes, but the grades are ordered and the decision is $P(\text{grade} \geq 2)$; four cumulative binary outputs would supervise that directly. (2) 224×224 makes one pixel roughly $50 \mu\text{m}$ while a microaneurysm is $15\text{--}60 \mu\text{m}$ — the earliest lesions are at the edge of what the input can carry; 512×512 needs a GPU. (3) IDRiD ships lesion masks, which would let the heatmap point at a haemorrhage instead of a region.

6. Domain and general-knowledge questions

Q17. What is diabetic retinopathy?

High blood sugar damages the tiny vessels at the back of the eye. They leak, close off, and the eye grows fragile new vessels that bleed.

In detail: It progresses through microaneurysms (grade 1), haemorrhages and exudates (grades 2–3), and finally new vessel growth — proliferative DR (grade 4) — which risks vitreous haemorrhage and retinal detachment. It is **painless and symptomless until late**, which is precisely why screening exists: by the time vision blurs, damage is already done.

Q18. How big is the problem in India?

Roughly 101 million people with diabetes (ICMR-INDIAB, 2023), around one in three of whom develop retinopathy — against roughly one ophthalmologist per 100,000 people.

In detail: Every diabetic is meant to have a dilated eye examination annually. The arithmetic does not work: the specialists do not exist in the numbers required, and they are concentrated in cities while most patients are not. DR is a leading cause of preventable blindness in working-age adults, and it is preventable precisely because early treatment works. The bottleneck is not knowledge — it is **who gets looked at, in what order**.

Q19. What is the treatment, and why does early detection matter so much?

Laser photocoagulation, anti-VEGF injections, or vitrectomy. Treatment prevents further loss but rarely restores vision already lost.

In detail: That asymmetry is the whole argument for screening. Catching grade 2 and treating it protects sight that still exists; catching grade 4 after a bleed means managing damage. This is also why a screening tool is tuned for sensitivity — the cost of the two error types is not symmetric.

Q20. Difference between screening and diagnosis?

Screening sorts a large healthy-looking population into "needs a closer look" and "does not". Diagnosis is the closer look, by a clinician.

In detail: A screening test is allowed to be imprecise in one direction: it should rarely miss disease, and may over-refer. That is the standard we designed to. RetiNova performs screening; a doctor performs the diagnosis, and every result carries that statement.

Q21. What is diabetic macular oedema, and why do you not detect it?

Fluid swelling at the macula, the part of the retina responsible for sharp central vision. We do not detect it because our labels do not contain it.

In detail: DME can blind someone at *any* retinopathy grade, so a "no DR" result does not rule it out — and our report says exactly that. Detecting it needs a dataset with macular oedema labels such as IDRiD's second task, or ideally OCT imaging, which a fundus camera cannot provide.

Q22. Why can a phone camera not photograph the retina?

The retina sits ~17 mm behind a 2–8 mm pupil. You must send light in and get the image back out through that same small opening without the cornea reflecting the light into the sensor.

In detail: Fundus cameras solve this with coaxial illumination and a condensing lens. A bare phone images the front of the eye and stops. That is why our retinal path needs a lens — from a ₹1,500 20-dioptre lens to a ₹15 lakh fundus camera — and why the camera-only path measures the person (pulse, pupil reflex, pallor) rather than the retina.

Q23. What do OD and OS mean?

Oculus dexter — right eye. *Oculus sinister* — left eye. (OU = both.)

In detail: Standard clinical notation, and recorded per screening. Laterality matters because the two eyes are graded and treated separately; a report that does not say which eye is clinically incomplete.

7. The SRS — requirements, in one page

An **SRS (Software Requirements Specification, IEEE 830 / ISO/IEC/IEEE 29148)** states what a system must do and how well. The short form:

Functional requirements

ID	Requirement	Status
FR-1	Patients and doctors register and sign in with separate roles	Done — JWT, BCrypt, role guards
FR-2	Capture a retinal image by camera or upload	Done — live capture and file upload
FR-3	Reject images that cannot be graded, with a reason	Done — 9 checks, ~15 ms, 11 tests
FR-4	Grade a retinal image on the 5-point ICDR scale	Done — EfficientNetB0 via ONNX
FR-5	Show which regions influenced the grade	Done — Grad-CAM heatmap + overlay
FR-6	Produce a referral decision at a stated operating point	Done — threshold on $P(\text{grade} \geq 2)$
FR-7	Route every screening to a doctor for review and sign-off	Done — live queue over SSE
FR-8	Produce a printable report with identity, grade, images, limitations	Done — print route + PDF
FR-9	Produce a priority score without any retinal hardware	Done — PPG, PLR, pallor
FR-10	Keep a per-patient history of screenings and triage	Done
FR-11	Let a doctor override the AI grade itself	Not done — decision and remarks only
FR-12	Compare a screening against the patient's previous ones	Not done — data is stored, nothing reads it

Non-functional requirements

ID	Requirement	Measured
NFR-1	Grade an image in under 3 seconds	13 ms model; 1–2 s end to end
NFR-2	Reject a non-eye fast enough for a live viewfinder	11–23 ms
NFR-3	Catch at least 80% of referable disease	80.9% internal, 95.4% IDRiD
NFR-4	Work on a mid-range phone browser without installation	Yes — HTTPS + getUserMedia
NFR-5	No patient sees another patient's data	Ownership checks on records and images
NFR-6	Run within free-tier resources	Yes — 16 MB model, CPU only
NFR-7	State its own error rates wherever it states a result	Yes — on screen and on the PDF

8. Standards, regulations and abbreviations

How deep does a judge go? Expect them to ask what an abbreviation stands for and why it matters — one or two sentences. They will rarely ask for clause numbers. The winning answer to "are you compliant?" is *not* "yes"; it is knowing which standard applies, and being straight that a hackathon prototype has not been certified against it.

Standards that apply to software like this

Standard	What it is	Where we stand
IEC 62304	Life cycle for medical device software — how it must be planned, built and maintained	Not followed formally. Would classify as Class B (non-life-threatening injury possible)
ISO 14971	Risk management for medical devices	Not formal, but the two main risks — ungradable images and false negatives — are identified and mitigated
ISO 13485	Quality management system for medical device manufacturers	Organisational, not applicable to a student project
ISO/IEC 27001	Information security management	Not certified. Auth and access control exist; audit logging and rate limiting do not
ISO/IEC/IEEE 29148	Requirements engineering — the SRS standard (supersedes IEEE 830)	Section 7 above follows its shape
DICOM	Digital Imaging and Communications in Medicine — the imaging format hospitals use	Not supported; we accept JPEG/PNG. Required for hospital PACS integration
HL7 FHIR	Fast Healthcare Interoperability Resources — how health records exchange data	Not implemented. The route to hospital and ABDM integration
ABDM / ABHA	Ayushman Bharat Digital Mission and its health account ID	Not integrated. The obvious path to national deployment in India
DPDP Act 2023	India's data protection law — consent, purpose limitation, erasure	Not addressed: no consent flow, no deletion path. Required before real patients
CDSCO / MDR 2017	India's medical device regulator and rules; software as a device is regulated	Would need registration as SaMD before clinical use
WCAG 2.1	Web accessibility guidelines	Partially — semantic HTML, keyboard navigation; contrast not audited

Abbreviations, by area

Term	Stands for	Means
Clinical		
DR	Diabetic Retinopathy	The disease being screened for
NPDR / PDR	Non-Proliferative / Proliferative DR	Grades 1–3 versus grade 4, the split at new-vessel growth
DME	Diabetic Macular Oedema	Swelling at the macula; separate cause of sight loss, not assessed here
VTDR	Vision-Threatening DR	Severe NPDR, PDR, or DME
ICDR	International Clinical Diabetic Retinopathy scale	The 0–4 grading our labels and report use
OD / OS / OU	Oculus Dexter / Sinister / Uterque	Right eye / left eye / both eyes
OCT	Optical Coherence Tomography	Cross-sectional retinal scan; the gold standard for DME
Anti-VEGF	Anti-Vascular Endothelial Growth Factor	Injected drug that suppresses abnormal vessel growth
Measurement		
Sensitivity	True positive rate	Of people with disease, the fraction we flag — ours is 80.9%
Specificity	True negative rate	Of healthy people, the fraction we clear — ours is 60.3%
PPV / NPV	Positive / Negative Predictive Value	Given a result, the chance it is right — depends on how common the disease is
QWK	Quadratic Weighted Kappa	Agreement with human graders, chance-corrected, distance-penalised
ROC / AUC	Receiver Operating Characteristic / Area Under Curve	The whole sensitivity-specificity trade-off in one curve and one number
Technical		
CNN	Convolutional Neural Network	The network family used for image recognition
GAP	Global Average Pooling	Averages each feature map to one number; what makes CAM equal Grad-CAM here
Grad-CAM	Gradient-weighted Class Activation Mapping	The heatmap technique
ONNX	Open Neural Network Exchange	Framework-independent model format; how we serve in 16 MB
JWT	JSON Web Token	Signed token proving who you are, without server-side sessions
SSE	Server-Sent Events	One-way live push; how the doctor's queue updates instantly
REST / API	Representational State Transfer	How the three services talk over HTTP
CORS	Cross-Origin Resource Sharing	Browser rule that lets our frontend call our API
PPG	Photoplethysmography	Pulse from light absorption changes in skin
PLR	Pupillary Light Reflex	How fast and how far the pupil constricts to light
SaMD	Software as a Medical Device	The regulatory category this product would fall into
SRS / HLD / LLD	Software Requirements Spec / High- / Low-Level Design	The three standard design documents
DFD / ERD / UML	Data Flow / Entity-Relationship Diagram / Unified Modeling Language	How the design is drawn

9. The three answers to have ready

"What is genuinely new here?" — Three things. It refuses images nobody could grade, before grading them. It works with no retinal hardware at all by measuring the person instead of the retina. And it decides referral at a stated operating point rather than by taking the model's favourite label — which is the difference between catching 65% and 81% of disease.

"What is the biggest weakness?" — Moderate (grade 2) recall is 20.5%, and that is exactly the referral boundary. The model separates obviously-healthy from obviously-diseased far better than it handles the middle. We know it, we measure it, and it is why a doctor reviews every case.

"Why should this win?" — Because it is measured rather than claimed. Every number in this document came from the running system, the failure modes are printed on the product itself, and it was validated on data from a different country and a different camera without being retrained to make the numbers look better.

RetiNova is a screening-support prototype, not a diagnostic device, and is not approved by any regulator. No output should be acted on without clinician review. Performance figures are measured on public research datasets and would require prospective validation on Indian screening populations before clinical deployment. Epidemiological figures are approximate and drawn from published sources (ICMR-INDIAB 2023 for diabetes prevalence).